Aufgabe 7.2

Setup

- Cores: 8
- Compiler/flags: `g++ -O2 -fopenmp`
- Scene: `test.stl`
- Resolutions: 512x512, 1024x1024
- Thread counts p: 1, 2, 4, 8

a) Runtimes T(p) and b) speedup S(p) / efficiency E(p)

Render only (excluding file I/O)

512x512

p	T(p) [s]	S(p)	E(p)
1	0.6939	1.00	1.00
2	0.3679	1.89	0.94
4	0.2488	2.79	0.70
8	0.1939	3.58	0.45

1024x1024

p	T(p) [s]	S(p)	E(p)
1	2.6378	1.00	1.00
2	1.5151	1.74	0.87
4	1.0081	2.62	0.65
8	1.1332	2.33	0.29

Total (including STL load + PPM write)

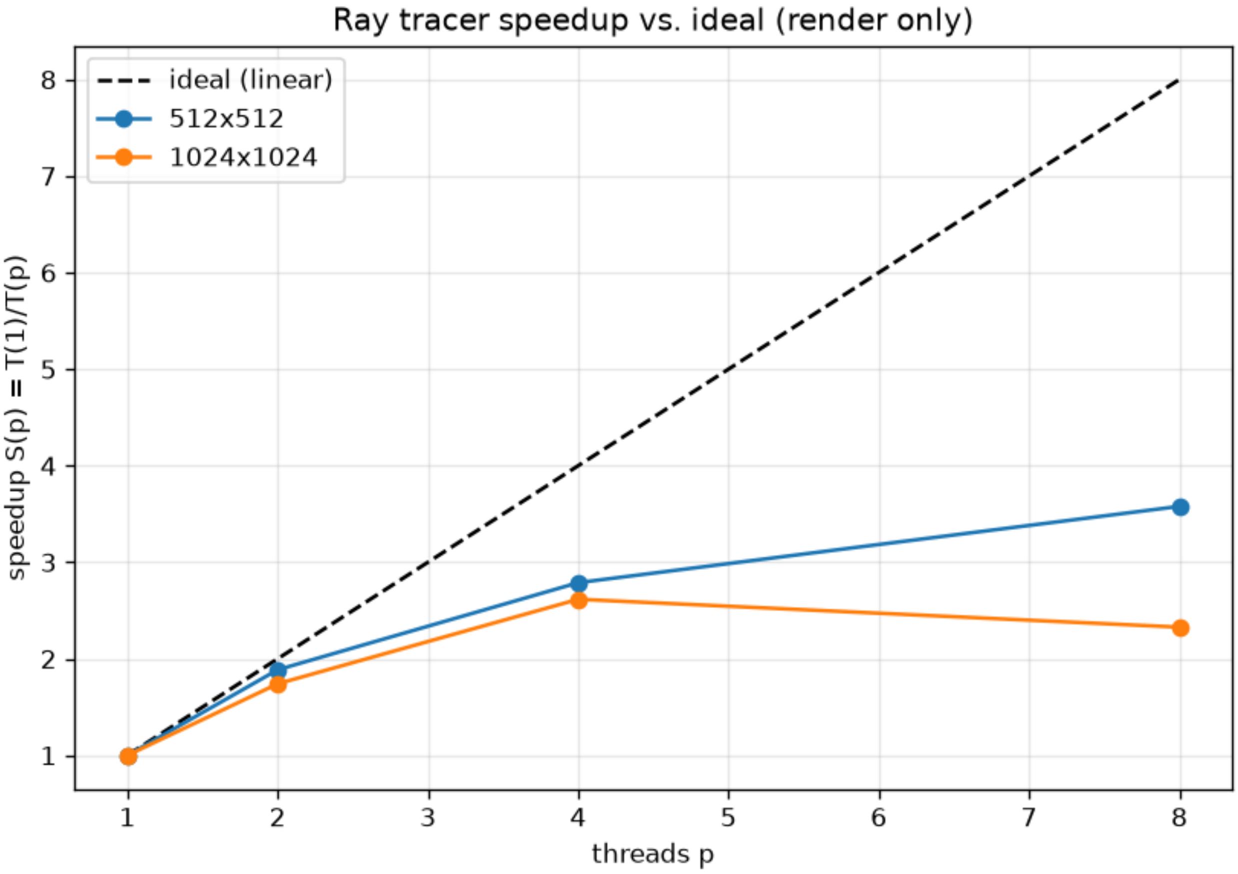
512x512

p	T(p) [s]	S(p)	E(p)
1	0.7223	1.00	1.00
2	0.4014	1.80	0.90
4	0.2830	2.55	0.64
8	0.2325	3.11	0.39

1024x1024

p	T(p) [s]	S(p)	E(p)
1	2.7640	1.00	1.00
2	1.6392	1.69	0.84
4	1.1498	2.40	0.60
8	1.3457	2.05	0.26

c) Speedup vs. ideal linear speedup



d) Amdahl's law fit

Fitting `S(p) = 1 / (f + (1-f)/p)` to the render-only speedups gives the serial fraction f (and an asymptotic speedup ceiling `1/f` as p -> infinity):

resolution	serial fraction f	speedup ceiling 1/f
512x512	0.1693	5.9
1024x1024	0.2907	3.4